Refractive Index

Exam code: 0625 & 0972

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Refractive index as a ratio of speed

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- The refractive index can be calculated in two different ways:

1. Using the **ratio of speeds**
2. Using the **ratio of angles**

- The refractive index is a number that is **always larger than 1** and is different for different materials
  - Objects which are more **optically dense** have a **higher** refractive index, e.g. *n* is about 2.4 for diamond
  - Objects which are less **optically dense** have a **lower** refractive index, e.g. *n* is about 1.5 for glass
- Since the refractive index is a **ratio**, it has **no units**
- The refractive index, *n*, for the ratio of speeds, is defined as:

The ratio of the speeds of a wave in two different regions

- The refractive index, *n*, for the ratio of speeds, is given by the equation:

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Got it

Refractive index as a ratio of angles

Extended tier only

- The refractive index, *n*, for the ratio of angles, is defined as:

The ratio of the sine of the angle of incidence and the sine of the angle of refraction of a wave in two different regions

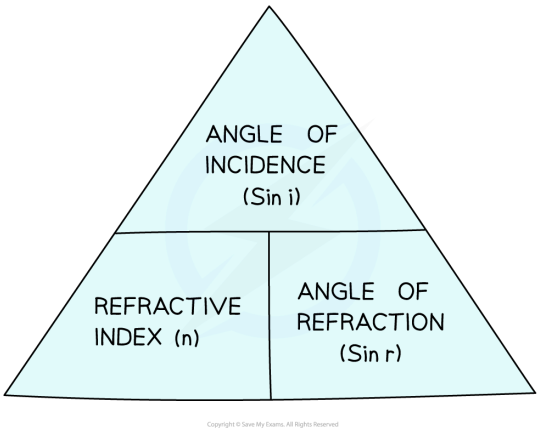
- The refractive index, *n*, for the ratio of angles, is given by the equation:

$$n = \frac{\sin i}{\sin r}$$

- Where:
  - *n* = the refractive index of the material
  - *i* = angle of incidence of the light (°)
  - *r* = angle of refraction of the light (°)

- This equation can be rearranged with the help of the formula triangle:

A refractive index formula triangle



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Formula triangle for the refractive index in terms of angles

- See the example of Speed & velocity which explains how to use the formula triangle to rearrange equations

Worked Example

A ray of light enters a glass block of refractive index 1.53 making an angle of 15° with the normal before entering the block.

Answer:

Step 1: List the known quantities

- Refractive index of glass, $n = 1.53$
- Angle of incidence, $i = 15°$

Step 2: Write the equation for refractive index in terms of the ratio of angles

$$n = \frac{\sin i}{\sin r}$$

Step 3: Rearrange the equation and calculate sin (r)

$$\sin r = \frac{\sin i}{n}$$

$$\sin r = \frac{\sin 15}{1.53}$$

$$\sin r = 0.1692$$

Step 4: Find the angle of refraction (*r*) by using the inverse sin function

$$r = \sin^{-1}(0.1692)$$

$$r = 9.7 = 10°$$

Examiner Tips and Tricks

$n = \frac{\sin i}{\sin r}$ is also known as Snell's law but you do not need to know this for your exam.

Important: $(\frac{\sin i}{\sin r})$ is not the same as $(\frac{i}{r})$. Incorrectly cancelling the sin terms is a very common mistake!

When calculating the value of *i* or *r* start by calculating the value of sin *i* or sin *r*.

You can then use the **inverse sin** function ($\sin^{-1}$ on most calculators by pressing 'shift' then 'sine') to find the angle.

One way to remember which way around *i* and *r* are in the fraction is remembering that '*i*' comes before '*r*' in the alphabet, and therefore is on the top of the fraction (whilst *r* is on the bottom).

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